PHYS 365 & 369 — Midterm Fall 2024 — Worked Solutions

20 questions. Every step uses the formula sheet printed at the back of the F2024 midterm. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key.

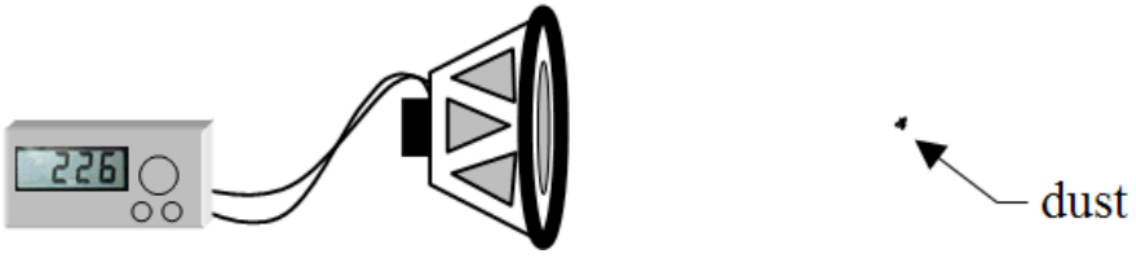
Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	A	11	B	16	C
2	B	7	A	12	C	17	A
3	A	8	C	13	C	18	A
4	A	9	A	14	C	19	A
5	A	10	A	15	A	20	A

REMEMBER — not on the formula sheet

- In a longitudinal wave the particles oscillate *along* the direction the wave travels, at the source frequency (Q1).
- Linear mass density $\mu = \dfrac{m}{L}$, and density $\rho = \dfrac{m}{V}$ (Q2, Q6, Q20).
- Compressing a cylinder along its length: $\dfrac{\Delta V}{V} = \dfrac{\Delta L}{L}$, because $V = \pi r^2 L$ and r stays fixed (Q6).
- A mass between two springs (one on each side): $k_{\text{eff}} = k_1 + k_2$, and the period is $T = \dfrac{2\pi}{\omega}$ (Q11).
- A hanging mass sets the tension: $\tau = Mg$ (Q20).
- In $I(r_2) = \left(\dfrac{r_1}{r_2}\right)^{n-1} I(r_1)\, e^{-\alpha \Delta r}$, n is the number of dimensions the sound spreads in. For a point source in air $n = 3$, which gives $\dfrac{1}{r^2}$ spreading (Q13).
- Log rules: $\log_{10}(ab) = \log_{10} a + \log_{10} b$ and $\log_{10}(e^x) = x \log_{10} e = 0.4343\, x$ (Q13).
- A taut string fixed at both ends uses the same formula as a pipe open at both ends, $f_n = \dfrac{nv}{2L}$. The F2024 sheet only labels it "pipe" (Q12, Q20).
- The harmonic number n is the number of loops (half-wavelengths) in the standing wave (Q19).

Question 1 Sound is a longitudinal wave

1. A particle of dust is floating in the air approximately half a meter in front of a speaker, which produces a pure tone of 226 Hz, as shown in the figure below. The sound waves produced by the speaker travel horizontally. Which one of the following statements correctly describes the subsequent motion of the dust particle?



A) It will oscillate left and right with a frequency of 226 Hz.
B) It will oscillate up and down with a frequency of 226 Hz.
C) It will accelerate to the right and continue moving in that direction.
D) It will remain motionless as it cannot be affected by sound waves.

Step 1. Sound waves in air are longitudinal. Each air particle (and the dust carried by the air) oscillates back and forth **along** the direction the wave travels, which here is horizontal (left–right).

Step 2. The particle is driven by the wave, so it oscillates at the wave's frequency, 226 Hz. It does not drift away with the wave. Only energy travels.

Final answer: **(A) Oscillates left and right at 226 Hz** ✓ Matches answer key

Question 2 Power carried by a string wave

2. A string is 2.70 m long, has a mass of 260 g, and is under a tension of 36.0 N. What must be the frequency of travelling waves of amplitude 7.70 mm for the average power to be 85.0 W?

A) 1241 Hz
B) 198 Hz
C) 1540 Hz
D) 343 Hz

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\tau}{\mu}}, \quad P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2, \quad \omega = 2\pi f$$

REMEMBER — not on the formula sheet

$$\mu = \frac{m}{L}$$

Step 1. Linear density and wave speed:

$$\mu = \frac{0.260}{2.70} = 0.09630\ \tfrac{\text{kg}}{\text{m}}, \qquad v = \sqrt{\frac{36.0}{0.09630}} = \sqrt{373.8} = 19.33\ \tfrac{\text{m}}{\text{s}}$$

Step 2. Solve the power formula for ω:

$$\omega = \sqrt{\frac{2P_{avg}}{\mu v\, y_m^2}} = \sqrt{\frac{2(85.0)}{(0.09630)(19.33)(7.70\times10^{-3})^2}} = \sqrt{\frac{170}{1.104\times10^{-4}}} = \sqrt{1.540\times10^{6}} = 1241\ \tfrac{\text{rad}}{\text{s}}$$

Step 3. Convert to frequency:

$$f = \frac{\omega}{2\pi} = \frac{1241}{2\pi} = 197.5\ \text{Hz} \approx 198\ \text{Hz}$$

(Option A, 1241, is ω in $\frac{\text{rad}}{\text{s}}$, not f. That is the trap.)

Final answer: **(B) 198 Hz** ✓ Matches answer key

Question 3 Superposing two equal waves

3. The following two waves travel along the same string (x is in meters and t is in seconds):

$$y_1 = (5.00\ \text{mm})\sin(4.00\pi x - 7.00\pi t)$$
$$y_2 = (5.00\ \text{mm})\sin(4.00\pi x - 7.00\pi t + 0.800\ \text{rad})$$

What is the amplitude of the resultant wave $y = y_1 + y_2$?

A) 9.21 mm
B) 4.65 mm
C) 3.49 mm
D) 6.98 mm

FROM THE FORMULA SHEET

$$y'(x,t) = \left[2y_m \cos\left(\tfrac{1}{2}\Phi\right)\right]\sin\left(kx - \omega t + \tfrac{1}{2}\Phi\right)$$

Step 1. Same amplitude $y_m = 5.00$ mm and phase difference $\Phi = 0.800$ rad. Make sure the calculator is in **radians**:

$$y'_m = 2y_m \cos\left(\frac{\Phi}{2}\right) = 2(5.00)\cos(0.400) = 10.0 \times 0.9211 = 9.21\ \text{mm}$$

Final answer: **(A) 9.21 mm** ✓ Matches answer key

Question 4 Tension and wave speed

4. The tension of a guitar string is increased by a factor of 4. How does the wave speed on the string change after this increase?

   A) The wave speed increases by a factor of 2.
   B) The wave speed decreases by a factor of 1/4.
   C) The wave speed decreases by a factor of 1/2.
   D) The wave speed increases by a factor of 4.

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\tau}{\mu}}$$

Step 1. Same string (same μ), so $v \propto \sqrt{\tau}$:

$$\frac{v'}{v} = \sqrt{\frac{4\tau}{\tau}} = \sqrt{4} = 2$$

Final answer: **(A) Speed increases by a factor of 2** ✓ Matches answer key

Question 5 Pressure amplitude vs. displacement amplitude

5. Which one of the following statements concerning the pressure amplitude of a traveling sound wave in air is **FALSE**?

   A) The pressure amplitude is inversely proportional to the displacement amplitude.
   B) The pressure amplitude is 90° out of phase with the displacement amplitude.
   C) A negative value of the pressure amplitude corresponds to an expansion of the air.
   D) When the displacement amplitude is at its maximum value, the pressure amplitude is zero Pascals.

FROM THE FORMULA SHEET

$$\Delta p_m = (v\rho\omega)\, s_m, \quad s = s_m \cos(kx - \omega t), \quad \Delta p = \Delta p_m \sin(kx - \omega t)$$

Step 1. $\Delta p_m = (v\rho\omega) s_m$ means Δp_m is **directly** proportional to s_m. Statement A says "inversely", so A is FALSE.

Step 2. Check the others. B: $\sin$ and $\cos$ are 90° apart, so B is true. C: a negative Δp is a pressure below normal, which is an expansion (rarefaction), so C is true. D: where $s = \pm s_m$ we have $\cos = \pm 1$, so $\sin = 0$ and $\Delta p = 0$, so D is true.

Final answer: **(A) "Inversely proportional" is false** ✓ Matches answer key

Question 6 Speed of sound in a liquid (bulk modulus)

6. A 20.0-cm-long, 10.0-cm-diameter cylinder with a piston at one end contains 1.34 kg of an unknown liquid. Using the piston to compress the length of the liquid by 1.00 mm increases the pressure by 4.15×10^6 Pa. What is the speed of sound in the liquid? [HINT: the volume of a cylinder is given by $V = \pi r^2 L$]

A) 986 m/s
B) 1973 m/s
C) 2206 m/s
D) 4932 m/s

FROM THE FORMULA SHEET

$B = -\dfrac{\Delta p}{\left(\frac{\Delta V}{V}\right)}, \quad v = \sqrt{\dfrac{B}{\rho}}, \quad V = \pi r^2 L$ (given as a hint)

REMEMBER — not on the formula sheet

$\rho = \dfrac{m}{V}$. At fixed radius, $\dfrac{\Delta V}{V} = \dfrac{\Delta L}{L}$.

Step 1. Volume and density, with $r = 0.0500$ m and $L = 0.200$ m:

$$V = \pi(0.0500)^2(0.200) = 1.571 \times 10^{-3}\ \text{m}^3, \qquad \rho = \frac{1.34}{1.571 \times 10^{-3}} = 853.1\ \tfrac{\text{kg}}{\text{m}^3}$$

Step 2. Fractional volume change:

$$\frac{\Delta V}{V} = \frac{\pi r^2\,\Delta L}{\pi r^2 L} = \frac{\Delta L}{L} = \frac{-1.00 \times 10^{-3}}{0.200} = -5.00 \times 10^{-3}$$

Step 3. Bulk modulus:

$$B = -\frac{4.15 \times 10^6}{-5.00 \times 10^{-3}} = 8.30 \times 10^8\ \text{Pa}$$

Step 4. Speed of sound:

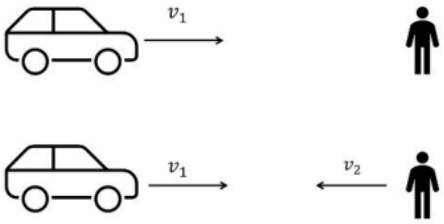
$$v = \sqrt{\frac{8.30 \times 10^8}{853.1}} = \sqrt{9.729 \times 10^5} = 986\ \tfrac{\text{m}}{\text{s}}$$

Final answer: **(A)** $986\ \frac{\text{m}}{\text{s}}$ ✓ **Matches answer key**

Question 7 Doppler, solving for the source frequency

7. A stationary person hears frequency f_1 from a source moving toward them at 20.0 m/s. While the source is still moving, the person then moves toward the source with a speed of 30.0 m/s and hears frequency f_2 that differs from f_1 by $f_2 - f_1 = 33.0$ Hz. What is the frequency f of the source? Take the speed of sound to be 340 m/s.

A) 352 Hz
B) 568 Hz
C) 389 Hz
D) 484 Hz



FROM THE FORMULA SHEET

$f' = f\dfrac{v \pm v_D}{v \pm v_S}$ (choose signs so that "toward" raises the frequency)

Step 1. Case 1: the person is stationary and the source moves toward them at 20.0:

$$f_1 = f\frac{v}{v - v_S} = f\frac{340}{320}$$

Step 2. Case 2: the person moves toward the source at 30.0 (numerator $+$) and the source still moves toward them:

$$f_2 = f\frac{v + v_D}{v - v_S} = f\frac{370}{320}$$

Step 3. Subtract:

$$f_2 - f_1 = f\frac{370 - 340}{320} = f\frac{30}{320} = 33.0 \;\Rightarrow\; f = 33.0 \cdot \frac{320}{30} = 352 \text{ Hz}$$

Final answer: **(A) 352 Hz** ✓ Matches answer key

Question 8 Phase difference from the resultant amplitude

8. The simple harmonic motion $y_s(t) = A\cos(\omega t)$ is the result of the superposition of two oscillations $y_1(t) = A\cos(\omega t + \varphi_1)$ and $y_2(t) = A\cos(\omega t + \varphi_2)$ of the same amplitude A. What is the phase difference $\Delta\varphi = |\varphi_2 - \varphi_1|$ between the two initial oscillations?

A) $\pi/3$
B) $3\pi/2$
C) $2\pi/3$
D) $\pi/4$

FROM THE FORMULA SHEET

$y' = \left[2y_m \cos\left(\frac{1}{2}\Phi\right)\right] \sin\left(\cdots + \frac{1}{2}\Phi\right)$ (the same algebra works for oscillations), $\cos(\alpha) + \cos(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$

Step 1. Add the two oscillations with the sum-to-product identity. The resultant amplitude is $2A\cos\left(\frac{\Delta\varphi}{2}\right)$.

Step 2. The resultant amplitude equals A:

$$2A\cos\left(\frac{\Delta\varphi}{2}\right) = A \;\Rightarrow\; \cos\left(\frac{\Delta\varphi}{2}\right) = \frac{1}{2} \;\Rightarrow\; \frac{\Delta\varphi}{2} = \frac{\pi}{3} \;\Rightarrow\; \Delta\varphi = \frac{2\pi}{3}$$

Final answer: **(C)** $\frac{2\pi}{3}$ ✓ **Matches answer key**

Question 9 Simple pendulum and mass

9. Two ideal pendulums have the same length but one of them has twice the mass of the other one. Which statement below regarding this situation is **TRUE**?

A) The two pendulums have the same frequency.
B) The pendulum with more mass has a shorter period.
C) The pendulum with more mass has a longer period.
D) The pendulum with the smaller mass has a smaller frequency.

FROM THE FORMULA SHEET

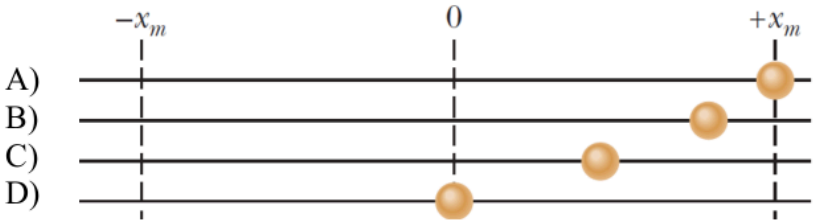
$\omega = \sqrt{\frac{g}{L}}, \quad T = 2\pi\sqrt{\frac{L}{g}}$

Step 1. Neither formula contains the mass. Same L and same g give the same ω, so the frequency and period are the same.

Final answer: **(A) Same frequency** ✓ **Matches answer key**

Question 10 Where is an SHM body at rest?

10. Select in which of the following positions of the simple harmonic motion with amplitude x_m the body performing the motion is at rest.



FROM THE FORMULA SHEET
$x(t) = x_m \cos(\omega t + \Phi)$

REMEMBER — not on the formula sheet
$v(t) = -\omega x_m \sin(\omega t + \Phi)$. Wherever $x = \pm x_m$ ($\cos = \pm 1$), $\sin = 0$, so $v = 0$.

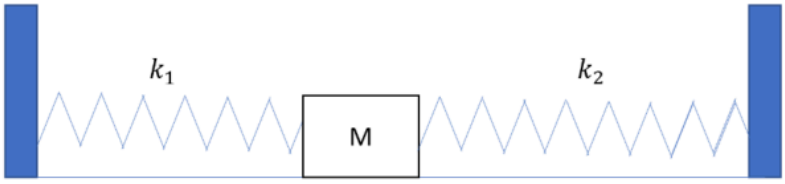
Step 1. The body is momentarily at rest at the turning points $x = \pm x_m$. In the picture only A has the ball at $+x_m$. At the centre (D) the speed is at its maximum.

Final answer: **(A) At $+x_m$** ✓ Matches answer key

Question 11 Mass between two springs

11. A body with mass of $M = 10.0$ kg is connected by two springs with respective spring constants $k_1 = 30.0$ N/m and $k_2 = 60.0$ N/m, as shown in the picture below. What is the period of the mass's simple harmonic motion?

A) 3.00 s
B) 2.09 s
C) 9.42 s
D) 15.1 s



FROM THE FORMULA SHEET
$\omega = \sqrt{\dfrac{k}{m}}, \quad \omega = \dfrac{2\pi}{T}$

REMEMBER — not on the formula sheet
Springs on opposite sides of a mass both push it back toward equilibrium, so $k_{\text{eff}} = k_1 + k_2$.

Step 1. Effective spring constant: $k_{\text{eff}} = 30 + 60 = 90 \ \frac{\text{N}}{\text{m}}$

Step 2. Angular frequency: $\omega = \sqrt{\dfrac{90}{10}} = \sqrt{9} = 3.0 \ \frac{\text{rad}}{\text{s}}$

Step 3. Period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3.0} = 2.09 \text{ s}$$

Final answer: **(B) 2.09 s** ✓ Matches answer key

Question 12 Tension for a guitar string

12. The High E string on the guitar has a linear mass density of $\mu_E = 3.09 \times 10^{-3}$ kg/m and is $L = 25.3$ in long. To be in tune, the string's fundamental frequency must be $f = 329.63$ Hz. What tension should be applied to the string? (Hint: 1.00 in = 2.54 cm)

A) 34.7 N
B) 850 kN
C) 554.6 N
D) 431.5 N

FROM THE FORMULA SHEET

$f = \dfrac{nv}{2L}$ (with $n = 1$ for the fundamental), $\quad v = \sqrt{\dfrac{\tau}{\mu}}$

Step 1. Convert the length: $L = 25.3 \text{ in} \times 2.54\frac{\text{cm}}{\text{in}} = 64.26 \text{ cm} = 0.6426 \text{ m}$

Step 2. Wave speed for the fundamental ($\lambda = 2L$):

$$v = 2Lf = 2(0.6426)(329.63) = 423.6 \ \tfrac{\text{m}}{\text{s}}$$

Step 3. Tension, from $v^2 = \dfrac{\tau}{\mu}$:

$$\tau = \mu v^2 = (3.09 \times 10^{-3})(423.6)^2 = (3.09 \times 10^{-3})(1.795 \times 10^{5}) = 554.6 \text{ N}$$

Final answer: **(C) 554.6 N** ✓ Matches answer key

Question 13 Sound level with spreading and absorption

13. The sound level at a distance 5.00 m away from a point source of sound is 90.0 dB. If the attenuation coefficient of the air is $\alpha = 1.74 \times 10^{-3}$ m^{-1}, what is the sound level at 100 m?

A) 64.7 dB
B) 25.3 dB
C) 63.3 dB
D) –26.7 dB

FROM THE FORMULA SHEET

$I(r_2) = \left(\dfrac{r_1}{r_2}\right)^{n-1} I(r_1)\, e^{-\alpha \Delta r}, \quad \Delta\beta = (10 \text{ dB}) \log_{10}\left(\dfrac{I_2}{I_1}\right)$

REMEMBER — not on the formula sheet

Point source: $n = 3$ (spherical spreading, $I \propto \dfrac{1}{r^2}$). Also $\log_{10}(e^x) = 0.4343\, x$.

Step 1. Intensity ratio, with $\Delta r = 100 - 5.00 = 95.0$ m:

$$\frac{I_2}{I_1} = \left(\frac{5.00}{100}\right)^2 e^{-(1.74\times10^{-3})(95.0)} = (0.0025)\, e^{-0.1653}$$

Step 2. Change in level:

$$\Delta\beta = 10\log_{10}(0.0025) + 10\log_{10}\left(e^{-0.1653}\right) = -26.02 + 10(-0.1653)(0.4343) = -26.02 - 0.72 = -26.74 \text{ dB}$$

Step 3. New level:

$$\beta_2 = 90.0 - 26.74 = 63.3 \text{ dB}$$

(Option A, 64.7 dB, leaves out the absorption term. Option D is just $\Delta\beta$.)

Final answer: **(C) 63.3 dB** ✓ Matches answer key

Question 14 Doppler, approaching vs. receding

14. An ambulance is passing you on the street. Due to the Doppler effect, the perceived frequency when the ambulance is approaching and when it is moving away have the following relation:

A) They are equal.
B) The perceived sound when the ambulance is approaching has a lower frequency.
C) The perceived sound when the ambulance is moving away has a lower frequency.
D) It depends on the wavelength of the sound when the source is not moving.

FROM THE FORMULA SHEET

$$f' = f\,\frac{v \pm v_D}{v \pm v_S}$$

Step 1. Approaching: $f' = f\,\dfrac{v}{v - v_S} > f$. Receding: $f' = f\,\dfrac{v}{v + v_S} < f$.

Step 2. So the pitch drops as the ambulance passes and moves away.

Final answer: **(C) Lower frequency when moving away** ✓ Matches answer key

Question 15 Designing sound insulation

15. You are given two materials. Material one has an impedance similar to that of air and a high attenuation coefficient. Material two has a very high impedance but a very low attenuation coefficient. Starting inside the room and going outward, how would you layer the two materials in order to make the sound insulation of a recording studio? (Hint: the studio must be insulated not only from outside sound but also from sound reflections inside.)

A) Material one, then material two, then the wall
B) Material two, material one, then the wall
C) The wall, then material two, then material one
D) The wall, then material one, then material two

FROM THE FORMULA SHEET

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad I(r_2) = \ldots e^{-\alpha \Delta r}$$

Step 1. **Inside surface**: material 1 has $Z \approx Z_{\text{air}}$, so $R \approx 0$. Sound from inside the room goes into it instead of echoing back, and its high α absorbs that sound. This removes reflections inside the room.

Step 2. **Next layer**: material 2 has a very large Z, so $|R| \approx 1$. Most outside sound is reflected before it can reach the inner layers. This blocks outside noise.

Step 3. Then the wall. The order from the inside going out is material 1, material 2, wall.

Final answer: **(A) Material one, material two, then the wall** ✓ Matches answer key

Question 16 Acoustic beam splitter

16. A beam splitter is a device which reflects half and transmits half of the amplitude of an incident light wave. Imagine a beam splitter constructed for sound waves. The impedance of air is $Z_1 = 413.3$ Pa s m^{-1}. What should the impedance of the material of the beam splitter be?

A) 206.65 Pa s m^{-1}
B) 1653.2 Pa s m^{-1}
C) 1239.9 Pa s m^{-1}
D) 137.76 Pa s m^{-1}

FROM THE FORMULA SHEET

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad T = \frac{2Z_1}{Z_1 + Z_2}$$

Step 1. We need $|R| = \frac{1}{2}$ **and** $T = \frac{1}{2}$. Write $r = \frac{Z_2}{Z_1}$.

Step 2. From $T = \frac{2}{1+r} = \frac{1}{2}$ we get $1 + r = 4$, so $r = 3$.

Step 3. Check R: $R = \frac{1-3}{1+3} = -\frac{1}{2}$, so $|R| = \frac{1}{2}$. ✓

$$Z_2 = 3Z_1 = 3(413.3) = 1239.9 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-1}$$

(Option D, $\frac{Z_1}{3}$, gives $R = +\frac{1}{2}$ but $T = 1.5$, so it fails the transmission condition.)

Final answer: **(C) 1239.9 Pa·s·m^{-1}** ✓ Matches answer key

Question 17 Tuning pan-flute pipes with beats

17. You are trying to tune a pair of pan flute pipes, which are air columns open at both ends. One of them is in tune at $f_1 = 261.63$ Hz. The other pipe is slightly too long and when played together they produce a sound with 6 beats per second. If the speed of sound in air is 343 m/s, how much should you cut from the wrong pipe to make it in tune?

A) 1.5 cm
B) 38 cm
C) 0.029 cm
D) 5.0 mm

FROM THE FORMULA SHEET

$f = \dfrac{nv}{2L}$ (open–open), $\quad f_{beat} = f_1 - f_2$

Step 1. The long pipe has the lower frequency: $f_2 = 261.63 - 6 = 255.63$ Hz.

Step 2. Lengths for the fundamental ($n = 1$):

$$L_1 = \frac{v}{2f_1} = \frac{343}{2(261.63)} = 0.6555 \text{ m}, \qquad L_2 = \frac{343}{2(255.63)} = 0.6709 \text{ m}$$

Step 3. Amount to cut:

$$\Delta L = 0.6709 - 0.6555 = 0.0154 \text{ m} \approx 1.5 \text{ cm}$$

Final answer: **(A) 1.5 cm** ✓ Matches answer key

Question 18 Identifying a pipe from its resonances

18. A pipe resonates in three successive resonant modes at frequencies 272 Hz, 340 Hz, and 408 Hz. The relevant speed of sound is 340 m/s. Based on this information, select the correct statement:

A) The pipe is open at both ends, and has length 2.50 m
B) The pipe is open at one end, closed at the other, and has length 1.25 m
C) The pipe is open at both ends, and has length 0.40 m
D) The pipe is open at one end, closed at the other, and has length 0.80 m

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FROM THE FORMULA SHEET

Open–open: $f = \dfrac{nv}{2L},\ n = 1, 2, 3\ldots$ Closed–open: $f = \dfrac{nv}{4L},\ n = 1, 3, 5\ldots$

Step 1. Gaps between successive modes: $340 - 272 = 68$ Hz and $408 - 340 = 68$ Hz.

Step 2. Test the open–open case (gap $= f_1 = 68$ Hz): $\dfrac{272}{68} = 4,\ \dfrac{340}{68} = 5,\ \dfrac{408}{68} = 6$. These are consecutive integers, so the pipe is open at both ends. (For a closed–open pipe we would need $f_1 = 34$ Hz, and then the ratios 8, 10, 12 are even, which is impossible.)

Step 3. Length:

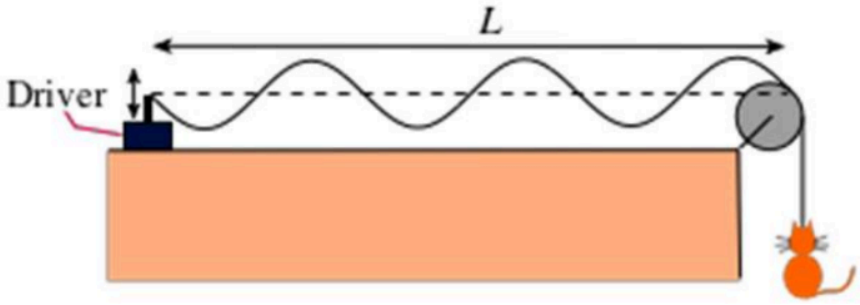
$$L = \frac{v}{2f_1} = \frac{340}{2(68)} = 2.50 \text{ m}$$

Final answer: **(A) Open at both ends,** $L = 2.50$ **m** ✓ **Matches answer key**

Question 19 Counting the harmonic number

Questions 19 and 20 are related to the following scenario:

A quirky scientist places a ceramic sculpture of a cat (mass $M = 3.00$ kg) on the end of a wire that passes over a massless, frictionless pulley, as shown in the figure below. The horizontal part of the wire has length $L = 0.750$ m, and this length of the wire has mass $m = 3.75$ g. A piston moves with simple harmonic motion and drives the left end of the wire such that a harmonic standing wave is created on the horizontal section, as shown in the figure. **Both ends of the string can be treated as if they are fixed.**



19. What is the value of the harmonic index n for the wave depicted?

A) 6
B) 7
C) 5
D) 4

FROM THE FORMULA SHEET

$f = \dfrac{nv}{2L}$, i.e. $L = n\dfrac{\lambda}{2}$

Step 1. Count the loops (half-wavelengths) between the driver and the pulley. The figure shows 3 full wavelengths, which is 6 loops.

$$L = 6 \cdot \frac{\lambda}{2} \Rightarrow n = 6$$

Final answer: **(A)** $n = 6$ ✓ **Matches answer key**

Question 20 Driver frequency

20. What is the frequency of the driver in hertz?

A) 307 Hz
B) 205 Hz
C) 256 Hz
D) 358 Hz

FROM THE FORMULA SHEET

$v = \sqrt{\dfrac{\tau}{\mu}}, \quad f = \dfrac{nv}{2L}, \quad g = 9.81\ \frac{\text{m}}{\text{s}^2}$

REMEMBER — not on the formula sheet

The hanging mass is in equilibrium, so $\tau = Mg$. Also $\mu = \dfrac{m}{L}$.

Step 1. Tension and linear density:

$$\tau = (3.00)(9.81) = 29.43\ \text{N}, \qquad \mu = \frac{3.75 \times 10^{-3}}{0.750} = 5.00 \times 10^{-3}\ \frac{\text{kg}}{\text{m}}$$

Step 2. Wave speed: $v = \sqrt{\dfrac{29.43}{5.00 \times 10^{-3}}} = \sqrt{5886} = 76.72\ \frac{\text{m}}{\text{s}}$

Step 3. Frequency for $n = 6$:

$$f = \frac{nv}{2L} = \frac{6(76.72)}{2(0.750)} = \frac{460.3}{1.50} = 307\ \text{Hz}$$

Final answer: **(A) 307 Hz** ✓ Matches answer key